Lotus bev4d数据说明汇总整理(标注商)

```
.
├── calib_anno
│   ├── camera_front_far.json
│   ├── camera_front_fisheye.json
│   ├── camera_front_wide.json
│   ├── camera_left_fisheye.json
│   ├── camera_left_front.json
│   ├── camera_left_rear.json
│   ├── camera_rear_fisheye.json
│   ├── camera_rear_mid.json
│   ├── camera_right_fisheye.json
│   ├── camera_right_front.json
│   └── camera_right_rear.json
├── calibration
│   ├── camera
│   ├── lidar
│   └── localization
├── logs
│   ├── peral-download.log
│   ├── peral-optpose.log
│   └── peral-timesync.log
├── node_output
│   ├── peral-publish
│   └── peral-timesync
└── sensor_data
    ├── camera
    ├── egopose
    ├── egopose.json
    ├── egopose_opt
    ├── fusion_object.json
    ├── gnss_rtk.json
    ├── imu.json
    ├── lidar
    ├── navi.json
    └── wheel_report.json

14 directories, 20 files
```

1. 概述

BEV 4D以一个数采包作为基本单元，包含激光、图像以及IMU等传感器的原始数据信息、传感器标定信息、以及运动补偿后的点云以及egopose等经过算法处理的信息。所有传感器的时间戳均在同一个时间域下进行授时，同时所有传感器的外参均指相对于车体坐标系的旋转平移关系，车体坐标系原点定义为车体后轴中心在地面的投影。

下面将分成传感器数据、标定文件、以及后处理信息三部分去介绍。

2. 传感器数据

2.1 去畸变顶上激光点云

- 激光数据以.pcd格式进行保存，
- 去畸变激光路径为：/sensor_data/lidar/lidar_undist
- 去畸变激光定位路径为: /sensor_data/egopose_opt/egopose_optpose
- 激光命名为"时间戳.pcd",
- 激光点包含信息：x, y, z, timestamp, intensity, ring, azimuth
- 激光点已经通过外参转到车体坐标系
- 时间戳单位均为ns

2.2 顶上激光拼接点云地图

- 激光数据以.pcd格式进行保存，
- 去畸变激光点云拼接路径为：/sensor_data/lidar/lidar_map
- 拼接点云定位路径为: /sensor_data/egopose_opt/egopose_pcd_map
- 激光命名为"时间戳.pcd",
- 激光点包含信息：x, y, z, timestamp, intensity, ring, azimuth
- 激光点已经通过外参转到第一帧车体坐标系
- 时间戳单位均为ns

2.3 毫米波雷达

- result_front:
  - 数据：/sensor_data/radar/result_front
  - 定位：/sensor_data/egpose_opt/result_front
- result_lc:
  - 数据：/sensor_data/radar/result_lc
  - 定位：/sensor_data/egpose_opt/result_lc
- result_lr:
  - 数据：/sensor_data/radar/result_lr
  - 定位：/sensor_data/egpose_opt/result_lr
- result_rc:
  - 数据：/sensor_data/radar/result_rc
  - 定位：/sensor_data/egpose_opt/result_rc
- result_rear:
  - 数据：/sensor_data/radar/result_rear
  - 定位：/sensor_data/egpose_opt/result_rear
- result_rr:
  - 数据：/sensor_data/radar/result_rr
  - 定位：/sensor_data/egpose_opt/result_rr

2.4 相机

11v相机的传感器数据包含7v的普通相机+4v的鱼眼相机，

- 相机数据以jpeg的格式保存，

- 数据路径/sensor_data/camera/camera_name
- 数据定位 /sensor_data/egopose_opt/camera_name
- 相机命名为"时间戳.jpeg", 时间戳指的是曝光结束时间

2.5 egopose_opt

自车后轮中心定位离线定位信息对应文件为/sensor_data/egopose_opt.json

```
struct MLALocalization {
  maf_std::Header header;    //!< message header
  MLALocalizationMeta meta; //!< meta data
  MLAPosition position;      //!< position (egopose:  global-llh, local-boot)
  MLAVelocity velocity;      //!< velocity (egopose:  global-llh, local-boot)
  MLAAngularVelocity angular_velocity; //!< angular velocity (local-car)
  MLAOrientation
      orientation; //!< orientation (egopose:  global-llh, local-boot)
  MLAAcceleration
      acceleration; //!< acceleration (egopose:  global-deprecated, local-car)
  MLAPositionStd position_std;                //!< variance of position
  MLAVelocityStd velocity_std;                //!< variance of velocity
  MLAAngularVelocityStd angular_velocity_std; //!< variance of angular velocity
  MLAOrientationStd orientation_std;          //!< variance of orientation
  MLAAccelerationStd acceleration_std;        //!< variance of acceleration
  MLAStatus status;                           //!< status info
  MLATransform transform;            //!< transform data for boot coordinate
  MLAPoseDetail pose_detail;         //!< pose detail for egomotion
  MLARoadLaneIdInfo road_lane_id_info; //!< road_id and lane_id infomation
  MLAReserved reserved;              //!< reserved
};

{
    "header": {
        "seq": 444630,
        "stamp": {
            "secs": 1693311537,
            "nsecs": 528672473
        },
        "frame_id": "{\"coordinate_system\" : \"GCJ02\", \"gnss_type\" : \"rtk\", \"loc_type\" : \"ddpf\", \"np_mode\" : \"hdmap\", \"version\" : \"1169e60c1c-HEAD\"}"
    },
    "meta": {
        "timestamp_us": 1693311537543940,  // 传感器时间
```

```
        "seq": 444630
    },
    "position": {
        "available": 3,
        "position_global": {
            "latitude": 27.91072136275722,          // gcj02坐标系
            "longitude": 112.88816407484823,
            "altitude": 0.0
        },
        "position_local": {
            "x": 2226.2792000572035,
            "y": -4803.918817306827,
            "z": -74.34943892907062
        }
    },
    "velocity": {
        "available": 3,
        "velocity_global": {
            "ve": 25.05373025113549,       // enu 下速度
            "vn": 8.558763080511905,
            "vu": 0.30843588537803357
        },
        "velocity_local": {
            "vx": -3.18864986702602,
            "vy": 26.383956670211738,
            "vz": 0.20710309885589742
        }
    },
    "angular_velocity": {
        "available": 1,
        "angvelocity_local": {
            "vx": -0.004916933694351262,
            "vy": 0.0031466404497138226,
            "vz": 0.004512908882351219
        }
    },
    "orientation": {
        "available": 15,
        "euler_global": {
            "roll": -0.01486402913256871,
            "pitch": -0.011856329782549146,
            "yaw": 0.32863650341342565
        },
        "quaternion_global": {
            "w": 0.9864927321304374,
            "x": -0.0063620141927859435,
            "y": -0.007063812294703622,
```

```
            "z": 0.163528947850529
        },
        "euler_local": {
            "roll": 0.005016836216351316,
            "pitch": -0.007839548346004118,
            "yaw": 1.6910068713193818
        },
        "quaternion_local": {
            "w": 0.6633548176526989,
            "x": 0.00459712678436107,
            "y": -0.0007231397248561343,
            "z": 0.7482905380874607
        }
    },
    "acceleration": {
        "available": 3,
        "acceleration_global": {
            "ae": -0.00288908346149117,
            "an": -0.005084340996984973,
            "au": 0.10538155635759744
        },
        "acceleration_local": {
            "ax": 0.012754958205008,
            "ay": 0.021476245489174373,
            "az": 0.9925713858018208
        }
    },
    "position_std": {
        "available": 2,
        "pos_std_global": {
            "std_pe": 0.0,
            "std_pn": 0.0,
            "std_pu": 0.0
        },
        "pos_std_local": {
            "std_px": 2.605007118698762,
            "std_py": 2.1440002492192005,
            "std_pz": 2.6992159362176236
        }
    },
    "velocity_std": {
        "available": 0,
        "vel_std_global": {
            "std_ve": 0.0,
            "std_vn": 0.0,
            "std_vu": 0.0
        },
```

```
        "vel_std_local": {
            "std_vx": 0.0,
            "std_vy": 0.0,
            "std_vz": 0.0
        }
    },
    "angular_velocity_std": {
        "available": 0,
        "angvel_std_local": {
            "std_vx": 0.0,
            "std_vy": 0.0,
            "std_vz": 0.0
        }
    },
    "orientation_std": {
        "available": 1,
        "ori_std_xyz": {
            "std_faix": 0.004728773086070612,
            "std_faiy": 0.00474757279414678,
            "std_faiz": 0.0017571112742306057
        }
    },
    "acceleration_std": {
        "available": 0,
        "acc_std_global": {
            "std_ae": 0.0,
            "std_an": 0.0,
            "std_au": 0.0
        },
        "acc_std_local": {
            "std_ax": 0.0,
            "std_ay": 0.0,
            "std_az": 0.0
        }
    },
    "status": {
        "available": 1,
        "status_info": {
            "quality": 0,
            "common": 3,
            "extended": 0,
            "type": 1
        }
    },
    "transform": {
        "available": 4,
        "transform_llh_to_boot": {
```

```
            "transform_q": {
                "w": 0.0,
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
            "transform_t": {
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
            "transform_center": {
                "latitude": 0.0,
                "longitude": 0.0,
                "altitude": 0.0
            }
        },
        "transform_avp_map_to_boot": {
            "transform_q": {
                "w": 0.0,
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
            "transform_t": {
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
            "transform_center": {
                "latitude": 0.0,
                "longitude": 0.0,
                "altitude": 0.0
            }
        },
        "transform_ego_motion_to_boot": {
            "transform_q": {
                "w": 1.0,
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
            "transform_t": {
                "x": 0.0,
                "y": 0.0,
                "z": 0.0
            },
```

```
            "transform_center": {
                "latitude": 0.0,
                "longitude": 0.0,
                "altitude": 0.0
            }
        }
    },
    "pose_detail": {
        "available": 0,
        "pose_detail_info": {
            "available": 0,
            "is_keyframe": false,
            "has_scale": true,
            "pose_type": 1
        }
    },
    "road_lane_id_info": {
        "ur_id": 0,
        "tile_id": 0,
        "road_count": 0,
        "lane_id": 0,
        "reserved": 0
    },
    "reserved": {
        "available": 3,
        "reserved_data": [
            0,
            0,
            0
        ]
    }
}
```

3. 标定参数

3.3 相机

路径：/calibration/camera/camera_name.yaml

相机标定的外参指的是相机到车体坐标系的外参，可以参考注释理解每个字段的含义。注意所有相机的camera_model都是等距相机模型

```
%YAML:1.0
---
CLOCK_calib_version: "1d987b7d05-dirty/master"
CLOCK_calib_details: "output by CLOCK calibration"
```

```yaml
CLOCK_calib_date: "2023-09-25-12:39:22"
sensor_name: "camera_front_wide"  # 传感器命名
sensor_type: "Camera"  # 传感器类型
timestamp_shift: 0  # 时间戳延时，单位 ms，真实时间戳 = 得到时间戳 + timestamp_shift
vehicle_xyz: "front_left_up"  # 车体系定义，前左上，后轴中心接地点
r_s2b: [-1.210487, 1.214899, -1.200436] # 传感器到车体系的旋转，i.e. p_b = R(r_s2b)*p_s，p_b表示车体系下的点，R(.)表示把旋转向量转换成旋转矩阵的函数，p_s表示传感器系的点
t_s2b: [2.1352317030037655, -0.00870754000000002, 1.526446] # 传感器到车体的平移，单位 m
camera_model: "polyn"  # 相机模型-poly，也叫等距相机模型
fx: 1911.701748  # 内参-焦距-fx，单位 像素
fy: 1911.701748  # 内参-焦距-fy，单位 像素
cx: 1917.431243  # 内参-主点-cx，单位 像素
cy: 1694.68327  # 内参-主点-cy，单位 像素
kc2: -0.034213  # 内参-畸变系数-kc2
kc3: -0.001692  # 内参-畸变系数-kc3
kc4: 0  # 内参-畸变系数-kc4
kc5: 0  # 内参-畸变系数-kc5
is_fisheye: false  # 是否是鱼眼相机
line_exposure_delay: 0  # 行曝光延迟，单位 us
width: 3840  # 图像宽度，单位 像素
height: 2160  # 图像高度，单位 像素
suggested_rect_region_within_ROI: [0, 0, 3840, 2160]  # 建议使用的图像ROI，单位 像素
suggested_diagonal_FOV_within_ROI: "N/A"  # 建议使用的ROI，单位 deg
```

4. 后处理信息

4.1 各传感器数据对齐

- 目录：/node_output/peral-dataproc/result.json
- 后续只会用到result.json中涉及到的传感器及其定位信息
- 注意事项：该json是bag包级别的传感器数据对齐，实际应用过程中可进行抽帧

4.2 单bag包拼接点云操作流程

名称	类型 / 大小
camera	目录
egopose	目录
egopose_opt	目录
lidar	目录
radar	目录
virtual_camera	目录
clockdiff.json	5.88KB
egopose.json	9.4MB
egopose_opt.json	1.83MB
fusion_object.json	10.11MB
gnss_rtk.json	679.29KB
imu.json	2.12MB
navi.json	17.68MB
weather.json	747B
wheel_report.json	3.71MB

- 拼接点云信息的处理

其中主要涉及lidar/lidar_map、egopose_opt/egopose_pcd_map/***.json和egopose_opt.json三个文件。

---

egopose_opt.json表示离线定位信息，包含整个bag包pcd时刻的定位信息，其中主要关注信息如下：

```
        "header": {
            "frame_id": "coordinate_system : GCJ02 version :df358d2",
            "seq": 691337,
            "stamp": {
                "nsecs": 50071040.0,      // 时间: 10^9 * secs + nsecs
                "secs": 1705916159
            }
        },
        "orientation": {
            "available": 15,
            "euler_global": {
                "pitch": -0.0014061722927710093,
                "roll": 0.020612624774028077,
                "yaw": -1.4835733251991803
            },
            "euler_local": {
```

```
                "pitch": -0.01561114301101936,
                "roll": 0.024410930808713576,
                "yaw": -0.040406288139229155
            },
            "quaternion_global": {
                "w": 0.7372282133130851,
                "x": 0.007123338732420298,
                "y": -0.007481215443341542,
                "z": -0.6755648384544521
            },
            "quaternion_local": {                // 旋转信息
                "w": 0.9996929225940114,
                "x": 0.012044626825169014,
                "y": -0.008049876414758277,
                "z": -0.02010440175169731
            }
        },
        "position": {
            "available": 3,
            "position_global": {
                "altitude": 0.0,
                "latitude": 30.19490913885456,
                "longitude": 120.21808021706185
            },
            "position_local": {                  // 位置信息
                "x": 1316.532791282615,
                "y": -455.05823799239715,
                "z": 384.1237404237604
            }
        },
```

其中lidar/lidar_map中包含拼接点云

egopose_opt/egopose_pcd_map/***.json包含拼接点云中各单独帧的定位信息，数据结构如下：

```
{
  "merge_frame": {                          // 拼接点云使用到的定位信息，表示为:时间.json
    "1": "1705916159050071040.json",    // 拼接点云的参考坐标系， 定义此帧位置为base_pose
    "2": "1705916159650044928.json",    // 通过时间在egopose_opt.json寻找正在的旋转与位置信息
    "3": "1705916160150087936.json",
    "4": "1705916160650119936.json",
```

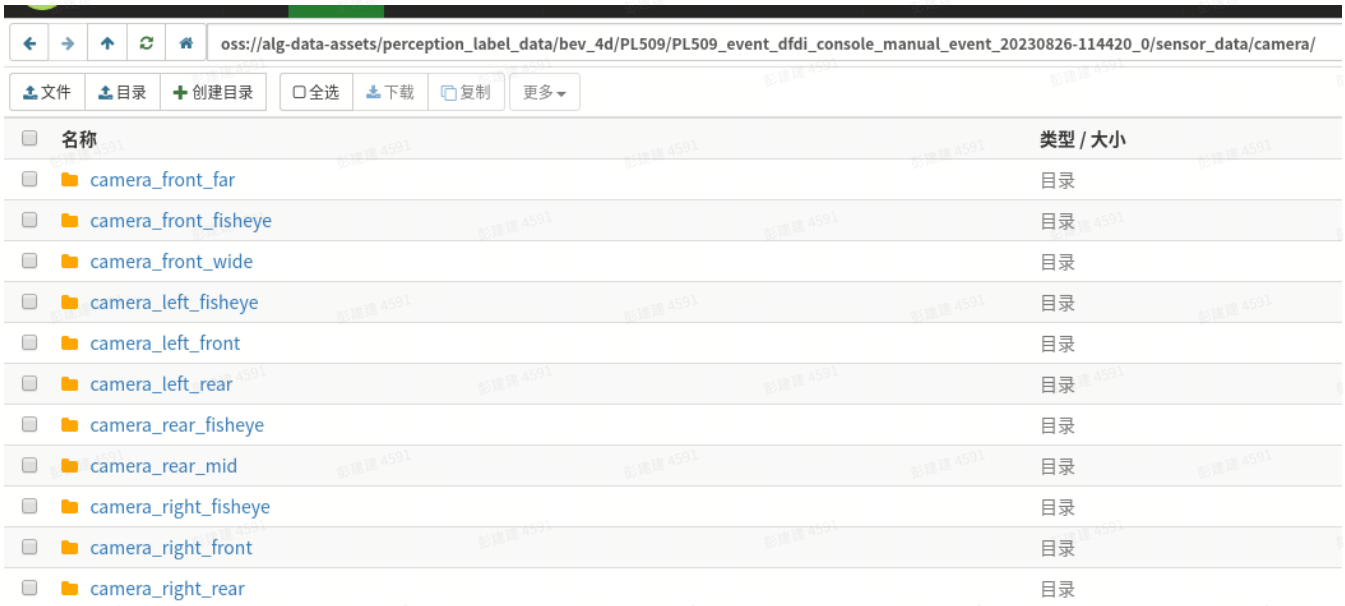
```
    "5": "1705916161250102016.json",
    "6": "1705916161750114048.json",
    "7": "1705916162250126080.json",
    "8": "1705916162850100992.json",
    "9": "1705916163350108928.json",
    "10": "1705916163950076928.json",
    "11": "1705916164450091008.json",
    "12": "1705916164950105088.json",
    "13": "1705916165550084096.json",
    "14": "1705916166050097920.json",
    "15": "1705916166550158080.json",
    "16": "1705916167050181888.json",
    "17": "1705916167650148096.json",
    "18": "1705916168250062080.json",
    "19": "1705916168750086912.json",
    "20": "1705916169250094080.json",
    "21": "1705916169850086912.json",
    "22": "1705916170350106880.json",
    "23": "1705916170850123008.json",
    "24": "1705916171450106880.json",
    "25": "1705916172050076928.json",
    "26": "1705916172550098944.json",
    "27": "1705916173150100992.json",
    "28": "1705916173750088960.json",
    "29": "1705916174250096896.json",
    "30": "1705916174850089984.json",
    "31": "1705916175450086912.json",
    "32": "1705916176050078976.json",
    "33": "1705916176550117888.json",
    "34": "1705916177050152960.json",
    "35": "1705916177650145024.json",
    "36": "1705916178250106112.json",
    "37": "1705916178850065920.json",
    "38": "1705916179450071040.json",
    "39": "1705916180050035968.json",
    "40": "1705916180650036992.json",
    "41": "1705916181150109952.json",
    "42": "1705916181650182912.json",
    "43": "1705916182150201088.json",
    "44": "1705916182750157056.json",
    "45": "1705916183250309888.json",
    "46": "1705916183850096896.json",
    "47": "1705916184450036992.json",
    "48": "1705916184950341888.json",
    "49": "1705916185450352896.json",
    "50": "1705916186050290944.json",
    "51": "1705916186650277120.json",
```

```
    "52": "1705916187250241024.json",
    "53": "1705916187850146048.json",
    "54": "1705916188450017024.json",
    "55": "1705916188950050048.json"
  }
}
```

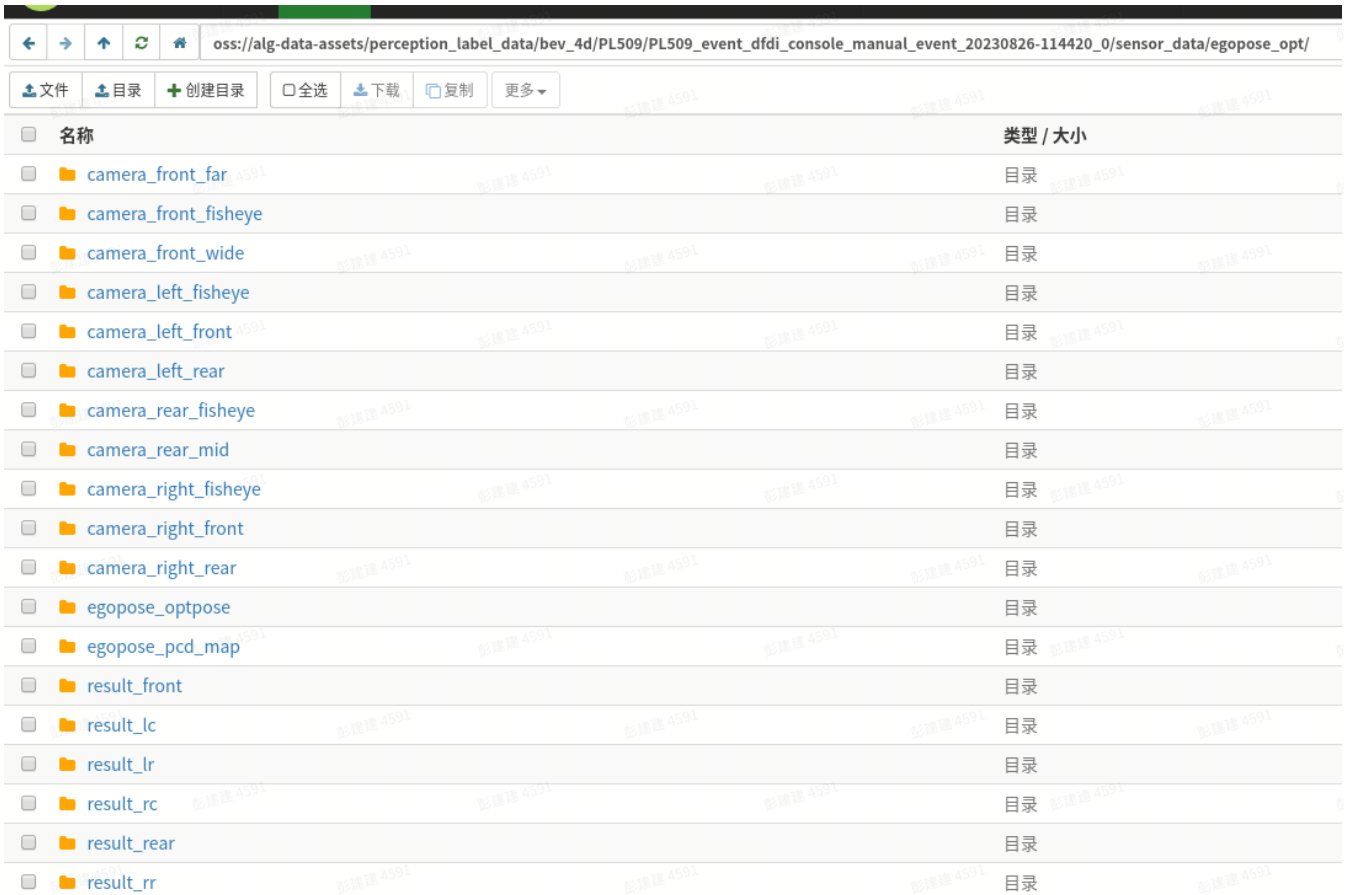
- 单帧图像数据处理

其中主要涉及到 camera_*.jpeg和相机定位camera_*.json

camera_*.jpeg路径为：



相机定位camera_*.json路径为：



定义：

camera_*.json 获取到的位姿为camera_pose

Pcd 标注成果为 p = [x, y, z]^T

相机外参为： camera_ex_pose

因此pcd 标注成果投影到图像物理坐标系为：

Pc = camera_ex_pose.inverse() * camera_pose.inverse() * **base_pose** * p

后续按照相机的内参矩阵K和畸变系数，实现图像投影，得到uv